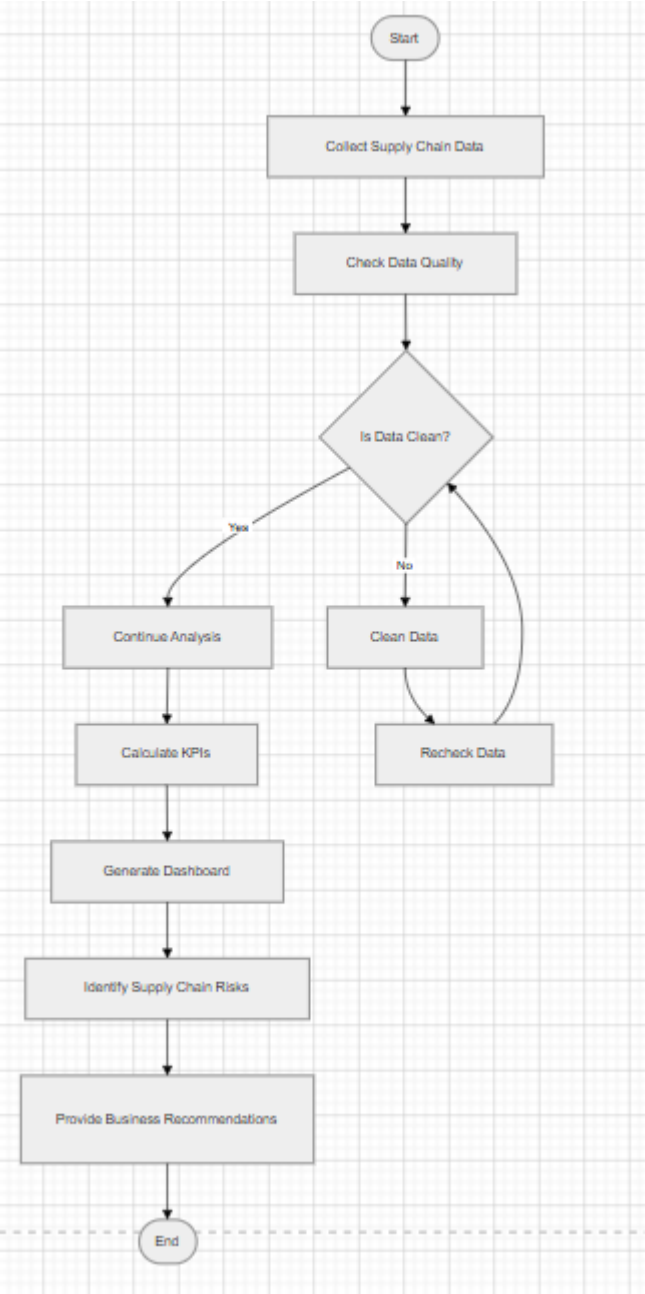
SUPPLY CHAIN PERFORMANCE AND INVENTORY OPTIMIZATION PROJECT

PROCESS FLOW DIAGRAM

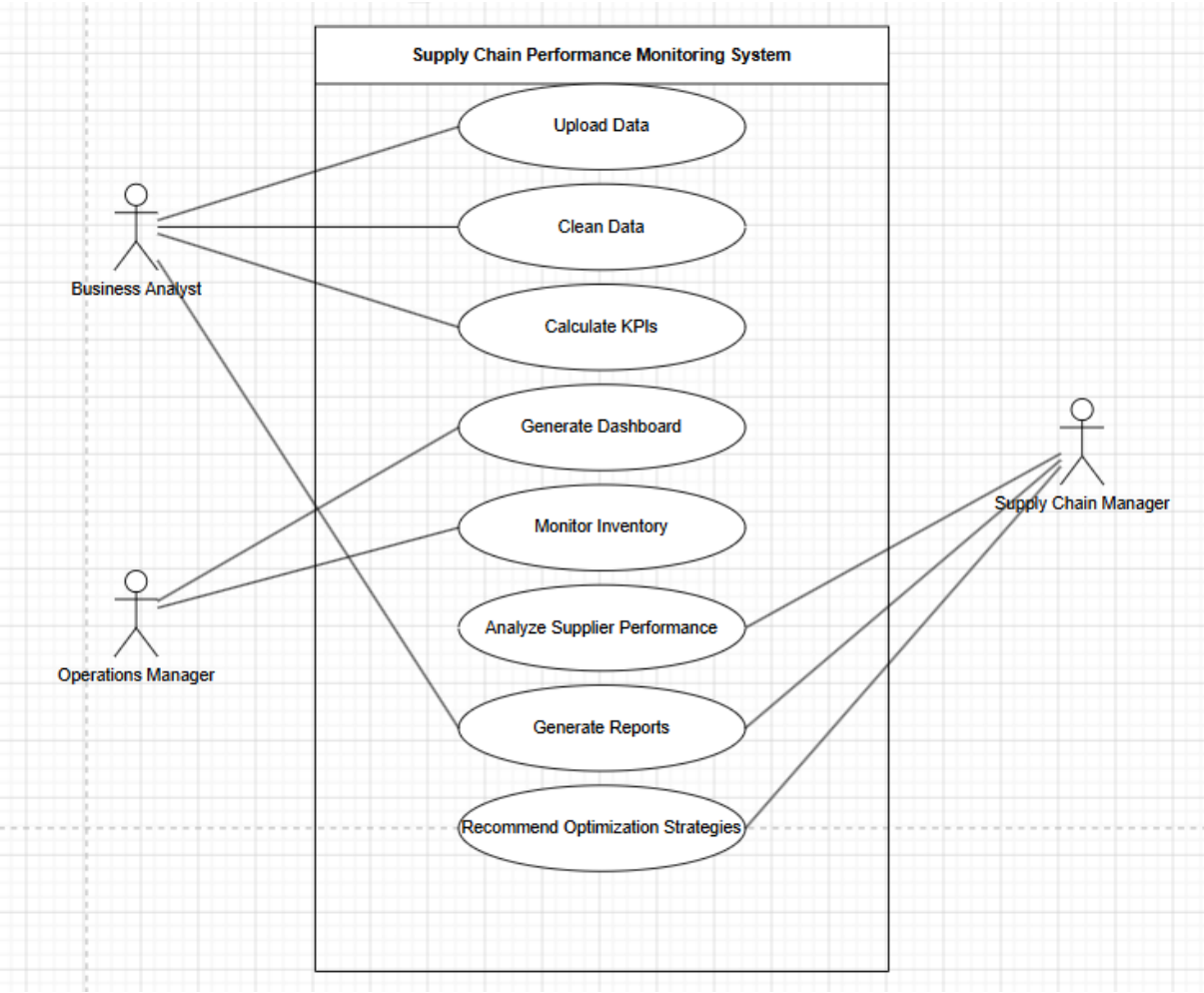


The supply chain process begins with collecting operational data from inventory and logistics systems. The data is cleaned and validated to ensure accuracy before calculating key performance indicators such as revenue, lead time, and defect rate. The cleaned data is then analyzed to identify supply chain risks and performance issues. A dashboard is generated to visualize trends and support decision-making. Based on the analysis, business recommendations are provided to improve inventory management and operational efficiency.

ACTIVITY FLOW DIAGRAM

The activity diagram represents the workflow involved in analyzing supply chain performance and supporting operational decisions. The Business Analyst begins by loading and validating the supply chain dataset. If data quality issues are identified, the data is cleaned and revalidated. Once the dataset is ready, key performance indicators are calculated and analyzed. The system generates a dashboard to visualize supply chain performance. The Operations Manager reviews the insights and approves improvement actions to optimize inventory and logistics operations.

USE CASE DIAGRAM



The use case diagram illustrates how different stakeholders interact with the supply chain analytics system to monitor inventory performance and support operational decisions. The Business Analyst is responsible for uploading, cleaning, and analyzing supply chain data to generate dashboards and reports. The Operations Manager uses the system to monitor inventory levels and review performance reports. The Supply Chain Manager analyzes supplier performance and implements improvement strategies based on analytical insights.

BUSINESS REQUIREMENTS DOCUMENT

Project Title: Supply Chain Performance & Inventory Optimization Analysis

1. Project Overview

The Supply Chain Performance & Inventory Optimization project aims to analyze operational data to improve inventory management, supplier performance, and logistics efficiency. The organization currently lacks centralized visibility into key supply chain metrics such as inventory levels, shipping time, defect rates, and supplier reliability. This project introduces a data-driven dashboard and reporting system to monitor supply chain performance and support informed business decisions.

The solution will enable stakeholders to identify operational inefficiencies, reduce supply chain risks, and optimize inventory management processes.

2. Business Problem Statement

The organization is experiencing challenges in managing inventory and monitoring supplier performance due to fragmented data and limited visibility into operational metrics. This results in:

- Overstocking and stock shortages
- Delayed shipments
- Increased operational costs
- Poor supplier performance tracking
- Inefficient decision-making

Management requires a centralized system to monitor supply chain performance and improve operational efficiency.

3. Project Objectives

The primary objectives of this project are:

- Improve visibility into supply chain operations
- Monitor inventory and supplier performance
- Identify operational risks and inefficiencies
- Reduce delays and inventory costs
- Support data-driven decision-making
- Enhance supply chain performance monitoring

4. Project Scope

In Scope

- Data cleaning and validation
- KPI calculation and analysis
- Dashboard development
- Supplier performance monitoring
- Inventory performance tracking
- Reporting and business recommendations

Out of Scope

- Real-time system integration
- Automated inventory ordering
- External supplier system integration
- Financial accounting processes

5. Stakeholders

Stakeholder	Role	Responsibility
Operations Manager	Decision Maker	Reviews supply chain performance
Supply Chain Manager	Process Owner	Monitors inventory and suppliers
Business Analyst	Analyst	Performs data analysis and reporting
Senior Management	Executive	Approves strategic decisions

6. Business Requirements

The system must support the following business requirements:

BR-1
The system shall provide visibility into inventory levels across products and locations.

BR-2
The system shall monitor supplier performance using key performance indicators.

BR-3
The system shall track shipping time and delivery performance.

BR-4
The system shall identify inventory shortages and overstock situations.

BR-5
The system shall generate reports and dashboards for supply chain monitoring.

BR-6
The system shall support data-driven decision-making for operational improvement.

7. Key Performance Indicators (KPIs)

The following KPIs will be used to evaluate supply chain performance:

- Total Revenue
- Total Inventory Value
- Average Lead Time
- Average Shipping Time
- Total Products Sold
- Defect Rate
- Supplier Performance
- Stock Availability

8. Assumptions

The following assumptions were made during the project:

- The dataset represents historical supply chain operations
- Data provided is accurate and complete after cleaning
- Stakeholders require periodic performance monitoring
- Supply chain operations follow standard logistics processes

9. Constraints

The following constraints were identified:

- Limited dataset size
- No real-time data integration
- Manual data updates
- Limited historical data

10. Risks

Potential risks associated with the project include:

- Inaccurate or incomplete data
- Supplier performance variability
- Operational delays
- Changes in demand patterns

11. Success Criteria

The project will be considered successful if:

- Supply chain performance metrics are clearly visible
- Inventory and supplier risks are identified
- Dashboard supports decision-making
- Business recommendations improve operational efficiency

Functional Requirements Document (FRD)

Project Title: Supply Chain Performance & Inventory Optimization Analysis

1. Introduction

The Functional Requirements Document (FRD) defines the functional capabilities and system behavior required to support supply chain performance monitoring and inventory optimization. The system will allow users to upload supply chain data, analyze key performance indicators (KPIs), monitor supplier performance, and generate dashboards for operational decision-making.

The purpose of this system is to provide stakeholders with a centralized platform to track inventory levels, shipping performance, and supplier efficiency.

2. System Overview

The Supply Chain Analytics System will process supply chain data and generate performance insights through dashboards and reports. The system will support data validation, KPI calculation, performance monitoring, and reporting functionalities to improve operational efficiency.

3. USER ROLES

User Role	Description
Business Analyst	Uploads data, performs analysis, and generates reports
Operations Manager	Reviews dashboard and monitors inventory performance
Supply Chain Manager	Monitors supplier performance and operational risks

4. Functional Requirements

FR-1 — Data Upload

The system shall allow the Business Analyst to upload supply chain data files in CSV or Excel format.

FR-2 — Data Validation

The system shall validate uploaded data to ensure completeness and accuracy.

FR-3 — Data Cleaning

The system shall allow the user to clean missing or incorrect data values.

FR-4 — KPI Calculation

The system shall calculate key performance indicators such as:

- Total Revenue
- Inventory Value

- Average Lead Time
- Average Shipping Time
- Defect Rate
- Products Sold

FR-5 — Dashboard Generation

The system shall generate interactive dashboards to visualize supply chain performance.

FR-6 — Inventory Monitoring

The system shall display stock levels for all products.

FR-7 — Supplier Performance Monitoring

The system shall monitor supplier performance using lead time and defect rate metrics.

FR-8 — Reporting

The system shall generate performance reports for operational review.

FR-9 — Data Filtering

The system shall allow users to filter data by:

- Product type
- Supplier name
- Location
- Shipping carrier

FR-10 — Risk Identification

The system shall identify operational risks such as:

- Low stock levels
- High defect rates and Delayed shipments

5. Input Requirements

The system will accept the following input data:

- Product type
- SKU
- Price

- Stock levels
- Supplier name
- Shipping carrier
- Lead time
- Shipping time
- Defect rate
- Revenue generated

6. Output Requirements

The system will generate the following outputs:

- KPI dashboard
- Inventory performance reports
- Supplier performance reports
- Shipping performance analysis
- Business recommendations

7. Non-Functional Requirements

Performance

The system shall generate reports within 5 seconds.

Usability

The dashboard shall be user-friendly and easy to navigate.

Reliability

The system shall ensure data accuracy after validation.

Security

Access to the system shall be restricted to authorized users.

Availability

The system shall be available during business hours.

8. Dependencies

The system depends on:

- Availability of accurate supply chain data

- User access to the dashboard platform
- Data updates from operational systems

User Stories

Project Title: Supply Chain Performance & Inventory Optimization Analysis

User Story 1 — Upload Data

US-1

As a **Business Analyst**,
I want to upload supply chain data into the system,
So that I can analyze operational performance.

User Story 2 — Clean Data

US-2

As a **Business Analyst**,
I want to clean missing or incorrect data values,
So that the analysis results are accurate and reliable.

User Story 3 — Calculate KPIs

US-3

As a **Business Analyst**,
I want to calculate key performance indicators,
So that I can measure supply chain performance.

User Story 4 — Monitor Inventory

US-4

As an **Operations Manager**,
I want to monitor stock levels,
So that I can prevent stock shortages and overstocking.

User Story 5 — Track Supplier Performance

US-5

As a **Supply Chain Manager**,
I want to track supplier performance,
So that I can identify unreliable suppliers.

User Story 6 — Identify Delays

US-6

As an **Operations Manager**,
I want to monitor shipping time,
So that I can identify delayed shipments.

User Story 7 — View Dashboard

US-7

As a **Manager**,
I want to view a dashboard of supply chain metrics,
So that I can make informed decisions.

User Story 8 — Generate Reports

US-8

As a **Business Analyst**,
I want to generate performance reports,
So that stakeholders can review operational results.

User Story 9 — Identify Risks

US-9

As a **Supply Chain Manager**,
I want to identify supply chain risks,
So that I can take preventive actions.

User Story 10 — Improve Operations

US-10

As a **Senior Manager**,
I want to receive operational recommendations,
So that I can improve supply chain efficiency.

3. Acceptance Criteria

User Story ID	Acceptance Criteria
US-1	User can upload CSV or Excel file successfully
US-2	Missing values are detected and corrected
US-3	KPIs are calculated correctly
US-4	Stock levels are displayed accurately

User Story ID	Acceptance Criteria
US-5	Supplier performance metrics are visible
US-6	Shipping delays are identified
US-7	Dashboard displays updated metrics
US-8	Reports are generated successfully
US-9	Risks are highlighted clearly
US-10	Recommendations are displayed to management

User Acceptance Testing (UAT)

Project Title: Supply Chain Performance & Inventory Optimization Analysis

1. UAT Objective

The objective of User Acceptance Testing (UAT) is to verify that the Supply Chain Performance Monitoring System meets business requirements and functions as expected. UAT ensures that stakeholders can successfully use the system to monitor inventory, analyze supplier performance, and support operational decision-making.

2. UAT Scope

The following system functionalities are included in UAT:

- Data upload and validation
- Data cleaning and processing

- KPI calculation
- Dashboard generation
- Inventory monitoring
- Supplier performance analysis
- Shipping performance tracking
- Report generation

3. UAT Participants

Role	Responsibility
Business Analyst	Conducts testing and documents results
Operations Manager	Validates inventory and logistics performance
Supply Chain Manager	Reviews supplier performance metrics
Project Manager	Approves final system acceptance

4. UAT Test Scenarios

Test Case ID	Test Scenario	Expected Result	Actual Result	Status
TC-1	Upload supply chain dataset	Data uploads successfully	Successful	Pass
TC-2	Clean missing values	Missing values are corrected	Successful	Pass
TC-3	Calculate KPIs	KPIs are calculated correctly	Successful	Pass
TC-4	Generate dashboard	Dashboard displays metrics	Successful	Pass
TC-5	Monitor inventory levels	Stock levels are visible	Successful	Pass
TC-6	Analyze supplier performance	Supplier metrics displayed	Successful	Pass

Test Case ID	Test Scenario	Expected Result	Actual Result	Status
TC-7	Track shipping performance	Shipping time displayed	Successful	Pass
TC-8	Generate reports	Reports generated correctly	Successful	Pass

5. UAT Acceptance Criteria

The system will be accepted if:

- All test cases pass successfully
- Dashboard displays accurate KPIs
- Reports generate without errors
- Users can monitor inventory and supplier performance
- System supports business decision-making

Business Insights and Recommendations

1. Inventory Optimization Recommendation

Insight

The analysis revealed that certain products maintain excessively high stock levels while others frequently experience low stock availability. This imbalance increases storage costs and creates a risk of stock shortages.

Recommendation

Implement inventory optimization strategies such as:

- Setting minimum and maximum stock thresholds
- Automating reorder alerts for low-stock products
- Reducing excess inventory for slow-moving items

Expected Business Impact

- Reduced inventory holding costs
- Improved product availability
- Better inventory turnover

2. Supplier Performance Improvement

Insight

Supplier performance analysis identified variations in lead time and defect rates across suppliers. Some suppliers consistently deliver late or provide lower-quality products.

Recommendation

Introduce supplier performance monitoring and evaluation processes:

- Establish supplier performance scorecards
- Monitor lead time and defect rate metrics regularly
- Replace or renegotiate contracts with underperforming suppliers

Expected Business Impact

- Improved product quality
- Reduced delivery delays
- Increased operational efficiency

3. Shipping Efficiency Optimization

Insight

Shipping performance analysis showed differences in delivery time across carriers. Some carriers take significantly longer to deliver products.

Recommendation

Optimize logistics operations by:

- Selecting faster and more reliable shipping carriers
- Monitoring shipping performance metrics
- Implementing delivery performance tracking

Expected Business Impact

- Faster deliveries
- Improved customer satisfaction
- Reduced operational delays

4. Demand Forecasting and Planning

Insight

Sales and inventory trends indicate fluctuations in product demand across locations and time periods.

Recommendation

Adopt demand forecasting techniques:

- Analyze historical sales trends
- Adjust inventory levels based on demand patterns
- Plan inventory replenishment schedules accordingly

Expected Business Impact

- Reduced stock shortages
- Improved inventory planning
- Better demand management

5. Risk Monitoring and Early Warning System

Insight

Operational risks such as delayed shipments, high defect rates, and low stock levels can disrupt supply chain performance.

Recommendation

Implement a risk monitoring system:

- Set alerts for high defect rates
- Monitor shipping delays
- Track inventory shortages

Expected Business Impact

- Early detection of supply chain issues
- Reduced operational risk
- Improved decision-making

Final Project Conclusion

Project Conclusion

The Supply Chain Performance & Inventory Optimization Analysis project successfully developed a data-driven framework for monitoring inventory and

supplier performance. By analyzing key performance indicators and visualizing operational metrics through dashboards, the project identified critical supply chain inefficiencies and operational risks.

The implementation of recommended strategies such as inventory optimization, supplier performance monitoring, and shipping efficiency improvements can significantly enhance operational efficiency, reduce costs, and support data-driven decision-making within the organization.